

The Market Game — Finding Equilibrium

ECON 2103 · Dr. Ke Lyu · classroom market experiment for 20–30 students

Students role-play a market. The instructor posts a price each round; buyers and sellers who want to trade at that price pair up and deal on the spot; the price then moves **toward balance** — up on a shortage, down on a surplus. Over a handful of rounds the price, and every trade, climbs to and settles at the equilibrium. Values are packed close to \$10 by design, so **almost every student trades** rather than a few sitting out. The winners are the sharpest traders: the buyer with the **lowest average price paid** and the seller with the **highest average price received**. A focused ~35-minute block.

Design at a glance

Original	This version
Buyers just have a money budget; sellers share one cost	Every buyer has a private <i>value</i>; every seller a private <i>cost</i> → real step-shaped demand & supply curves and an exact equilibrium price
“Raise or lower the price” (unspecified)	A proportional rule: $\text{next price} = \text{price} + k \times (\text{Demand} - \text{Supply})$ — the price steps toward balance each round and settles when Demand = Supply
A few extreme cards never trade	Values packed near \$10 so 14 of 15 on each side trade; the lone marginal card per side can be handed a unit by the dashboard's draw
Winner = lowest average price / highest profit	Kept and made central: buyers compete on lowest average price paid , sellers on highest average price received — a transparent test of <i>timing skill</i>
Trades vaguely first-come-first-served	Trades happen every round: willing buyers and sellers pair up at the posted price, so prices (and your average) are earned round by round
Instructor tallies and does the math by hand	A projected dashboard counts, applies the rule, tracks the market's average price, and draws the

Original	This version
	converging curves live

The market (exact numbers)

15 **buyers** and 15 **sellers**. Each card carries a private number, and each student has **3 units to trade** over the game (at most one per round).

Buyer values (max they'll pay per unit): 14, 13, 13, 12, 12, 12, 11, 11, 11, 11, 10, 10, 10, 10, 9 → cards **B1 ... B15**

Seller costs (what a unit costs them to make): 6, 7, 7, 8, 8, 8, 9, 9, 9, 9, 10, 10, 10, 10, 11 → cards **S1 ... S15**

Every value sits within a few dollars of \$10 by design, so **almost the whole room trades**. At the equilibrium price **\$10**, 14 of the 15 buyers (values ≥ 10) still profit and 14 of the 15 sellers (costs ≤ 10) still profit. Only the single most-marginal card on each side (buyer **B15**, value 9; seller **S15**, cost 11) is priced out at exactly \$10 — and you can still hand each of them a unit with the dashboard's draw during a \$9 or \$11 round. Above \$10 there's a surplus, below it a shortage, so the price is pulled to \$10, where the average trade price settles. (The dashboard shows this live.)

Class-size flexibility (20–30 students)

Cards are ordered so distribution is trivial. Let **N = half your class** (10–15 for 20–30 students). Hand out **B1...B_N** and **S1...S_N** — always matched numbers. The equilibrium price stays right around **\$10** for any N in this range. Odd class? Give the extra student a card and note the market leans that way slightly.

How a round works

1. **Instructor posts a price** on the dashboard. Round 1 starts at **\$6** — below everyone's expectations, so it's a shortage and the price has to climb (early units are cheap for whoever grabs them).
2. **Decide, then reveal**. You may trade **one unit this round** if it's worth it and you have a unit left:
 - **Buyers** raise their card only if the price is $\leq$ **your value** and you choose to buy now.

- **Sellers** raise only if the price is $\geq$ **your cost** *and* you choose to sell now.
 - You're allowed to **hold out** for a better price — but units get scarce when the price is far from normal, so waiting is a gamble.
3. **Count** the raised cards → **Demand** and **Supply** (enter both in the dashboard).
 4. **Trade**. Buyers and sellers **pair up** and deal at the posted price. The number of deals is **min(Demand, Supply)** — the short side all trade; whoever's on the long side and can't find a partner keeps their card down and tries next round. Everyone who traded **writes the price on their card** and crosses off a unit.
 5. **Adjust the price** (the rule below) and run the next round. Stop when almost no one wants to trade — the price has found its level.

The price rule — how surplus/shortage moves the price

The price change is **proportional to the size of the imbalance**:

$$\text{Next price} = \text{current price} + k \times (\text{Demand} - \text{Supply})$$

- **Shortage** (Demand > Supply) → the term is positive → **price rises**, harder the bigger the shortage.
- **Surplus** (Supply > Demand) → negative → **price falls**, more steeply the larger the glut.
- **Balanced** (Demand = Supply) → zero → the price holds. It has found the **equilibrium**.

A gentle climb that includes everyone. Because the values are packed close to \$10, the price only needs to travel a few dollars, so it climbs a step at a time and settles cleanly: a typical path from the \$6 start is **\$6 → \$8 → \$9 → \$10**. It visits the in-between prices, which is exactly what lets the lower-value buyers grab a cheap unit at \$6–\$8 and gives every card a turn. (A packed market has steep curves, so a big, twitchy step size would overshoot and never settle — this design keeps the steps small on purpose.)

Sensitivity control (k). The dashboard's toggle sets the step size: **Gentle (k = 0.15, default)** climbs one dollar at a time; **Quick (k = 0.25)** takes slightly bigger steps if you're short on time. Keep it on **Gentle** for the fullest participation, since more price stops mean more chances for the marginal cards to trade.

Scoring — who wins

Because trades happen **every round at that round's price**, each trader ends up with up to three prices on their card. Score = the **average** of your prices.

- **Top Buyer = the lowest average price paid.** You win by grabbing cheap units early, when the price is still below \$10 — but those rounds have few willing sellers, so it takes nerve and quick pairing.
- **Top Seller = the highest average price received.** You win by holding your units until the price has climbed — but wait too long and you may not find a buyer.

It's a clean, public test of timing: everyone sees the same prices, and the metric is simply “what did you pay / get, on average.” (The theoretical benchmark is \$10 — beat it and you outraded the market.) Poll the class at the end: buyers call out their average, lowest hand wins; then sellers, highest hand wins.

*The dashboard also tracks the **market's volume-weighted average price** — watch it converge toward \$10 even though individual trades happen at \$6, \$8, \$9, \$10 along the way. That convergence is the headline result of real market experiments.*

Timing (a ~35-minute block)

Segment	Time
Explain roles + rules (rules slides)	6–7 min
Hand out cards, one practice round	4–5 min
Trade: climb to \$10 and clear (~7 rounds with 3 units)	14–18 min
Poll the winners + debrief the curves and the average price	6–8 min

From the \$6 start the price climbs to equilibrium (\$6 → \$8 → \$9 → \$10) in about 3–4 rounds, then holds at \$10 for another 2–3 rounds while everyone clears their remaining units — roughly **7 rounds** with 3 units each, and nearly the whole room trades every round. Lean into the debrief (the converging curves, the winners, why the one marginal card on each side sits out) to round out the block.

Want more rounds? Give each student **more units** — the clean lever that lengthens the clearing phase: **5 units** → **~9 rounds**, **6** → **~11**. (The cards ship with a 3-unit log; ask if you'd

like a 5-unit version.)

Materials in this folder

- **Market_Game_Cards.pdf** — 30 printable cut-out cards (15 buyers, 15 sellers), each with its private number, the trade rule, and a 3-line log for recording your prices.
- **Market_Game_Rules_Slides.pdf** — projectable deck to explain the rules.
- **Market_Game_Instructor_Guide.pdf** — this design as a one-stop run sheet + answer key.
- **Market_Game_Dashboard.html** — the projected game runner (also a shareable web link).

Recommendation on tech: a purpose-built web dashboard beats Quizizz/Kahoot here. Quiz tools are built for “everyone answers the same question and gets scored” — they can’t tally a demand-vs-supply count, apply a price-adjustment rule, track a running average price, or draw converging curves. The dashboard does exactly that; students need no devices, and you keep the energy of everyone raising a card and pairing up to trade.